

Grid and shake: spatial aggregation and the robustness of regionally estimated elasticities

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Abstract This paper proposes a simple and transparent method for measuring spatial robustness of regionally estimated coefficients and considers the role of the administrative districts and of the size of regions. The procedure offers a new solution for a practical empirical issue: comparing the variables of interest across spatially aggregated units. It improves upon existing methods, especially when spatial units are heterogeneous. To illustrate the method, we use Hungarian data and compare estimates of agglomeration externalities at various levels of aggregation. Using the procedure, we find that the method of spatial aggregation seems to be of equal importance to the specification of the econometric model.

JEL Classification R12 · R30 · C15

1 Introduction

The concentration of economic activity is a key feature of economic geography; it is a well-researched topic in economics, and the polarization of activity is often a policy concern. As a result, measuring the agglomeration elasticity of market outcome

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variables such as wages, firm dynamics, or competition is of substantial importance.¹ Oftentimes, these studies are based on an arbitrary choice of spatial unit—cities, counties or regions—and shall be hence interested in the robustness of their results given the choice of unit. Furthermore, we may be interested in comparing results obtained on one country to estimates from another.

Hence, we need to understand how the specific characteristics of selecting the features of spatial units may affect the outcome of an estimation exercise. This paper will present a method to transparently investigate the impact of modeling choices of spatial aggregation that can be applied in any spatial setting with different layers of aggregation. The method will allow us to compare the magnitude of differences to aggregation to those resulting from running different regression with different controls.²

The unit of spatial scope can make a difference since externalities are often found to be in operation but fading away at a distance. For instance, [Fotheringham and Wong \(1991\)](#) argued that the features of areal units affect regression results—often in an uncertain fashion—while [Dewhurst and McCann \(2007\)](#) supplied evidence that the relationship between specialization and urban structure is a function of the scale of analysis. Using UK data on administrative units, they found a strong negative relationship between regional specialization and regional size. [Burger et al. \(2010\)](#) found different effects of agglomeration forces across geographic levels in the Netherlands. More recently, [Larsson \(2014\)](#) investigated how the wage density elasticity depends on spatial resolution in Swedish city areas and highlighted how quickly agglomeration effects attenuate with distance. Furthermore, the size of observation units and some of their features may be correlated and hence, estimates may be biased. For instance, large municipalities may be clustered in space because of first geography, specialization in agriculture, hard barriers to migration, distance to markets—crucial aspects of spatial economic structure that we may be looking at. Furthermore, the spatial scope of an activity may or may not coincide with the spatial scope of a political-administrative unit. At the same time, policies may affect relationships wherever interacted. Hence, any measurement at some regional level should combine the role of size (inclusion of externalities within) as well as the possible impact of regional policies.

Consequently, there is a need to compare in a statistically meaningful way estimation results across spatial units of different sizes taking into account the uneven (non-random) distribution of unit size. To this end, this paper offers a transparent, easy-to-use method.

The core idea of our procedure is to set up an artificial grid over the map of the country, assign municipalities (or ZIP codes) to it and then randomize the position of the grid to get a distribution of results. To do this, first, we create a grid of squares of a given size, L . Second, we take this grid of $L \times L$ squares and randomize its place on a map and, as a result, obtain a set of random realizations of artificial spatial units. By repeating this

¹ See, for instance, [Ciccone and Hall \(1996\)](#) on the relationship between labor productivity and agglomeration, [Ács and Armington \(2004\)](#) on entrepreneurship. Examples on local aspect of activity include [Iyer and Seetharaman \(2008\)](#) on retail prices, [Hortaçsu and Syverson \(2007\)](#) on local competition and vertical integration, [Gobillon and Milcent \(2013\)](#) on hospital performance.

² Note that our questions are closely related to that of [Amrhein \(1995\)](#) who argued that aggregation may be indeed an important methodological choice.

procedure (in this paper 300 times) we can get smooth kernel distributions. Third, we iron out the border effect by merging very small areas into bigger ones. Finally, we run our regression on each of these realizations and gather coefficients that yield a distribution of our parameters of interest. After this procedure, we can look at moments of this distribution, relate the distribution of coefficients to a single measure, for example, at the administrative regional unit, or compare distributions from two randomizations.

Our method is related to several earlier approaches proposed to treat the spatial aspects of estimations. The treatment of the location issue will often include spatial regression methods, and spatial lags in various forms have been widely used to gauge uncertainty about the right spatial scope (Anselin 2010). Artificial aggregation methods were also put forward. Amrhein and Flowerdew (1992) studied Canadian migration data both along region size and created artificial regions by setting a seed and randomly adding neighbors. In a close exercise to ours, Briant et al. (2010) placed starting points randomly on the map and added up an equal number of municipalities until the country was fully covered. Also closely related to this work is Menon (2012) who studied concentration patterns of US industries to disentangle labor market forces and set up a counter-factual exercise.³

We will be arguing that our method has the following merits: (i) it allows for carrying out estimations at various levels of aggregation and compare evidence; (ii) it treats uneven and non-random distribution of administrative unit size; (iii) it has the ability to compare results on administrative and artificial units, and (iv) it is able to gauge the statistical significance of differences. We will show that existing methods can match some but not all of these merits.

To illustrate our procedure, we will use Hungarian data—see an illustrative map of Fig. 1. Hungary is a country of 10 million inhabitants, over 93 thousand square kilometers, with 3125 municipalities. There are 150 micro-regions (*kistérség*), 20 NUTS3 (*megye*) and seven NUTS2 regions defined—see details in Table 3 of “Appendix” section. At the time of our empirical exercises, administrative duties had been shared by municipalities, micro-regions and NUTS3 regions. Briant et al. (2010) argued (p 301) that for historical reasons yielding very similar regions, the French data are well suited for their approach. We believe that for historical and geographic reasons, the Hungarian landscape is rather different and its unbalanced size distribution offers a laboratory to study our method of spatial aggregation and allow for a comparison with other approaches, such as the one by Briant et al. (2010).

We will consider two examples, estimating the agglomeration premium of wages and new firm formation, and we compare elasticities at two administrative units, i.e., municipal (NUTS5) and micro-region (NUTS4) levels, and for three artificial units. We are interested in the relationship between an estimated coefficient at an administrative regional level and a comparable estimate for an artificial region of similar size as well as how the average size of the defined area affects the estimated coefficient. Further, our method will be compared to earlier approaches and we argue that our methodology is better able to capture some important real-life features such as the non-random

³ There is a growing amount of related literature on a different type of setup, when the data are geocoded. For instance, Dubé and Brunelle (2014) argued that distance-based concentration measures may offer a good solution.

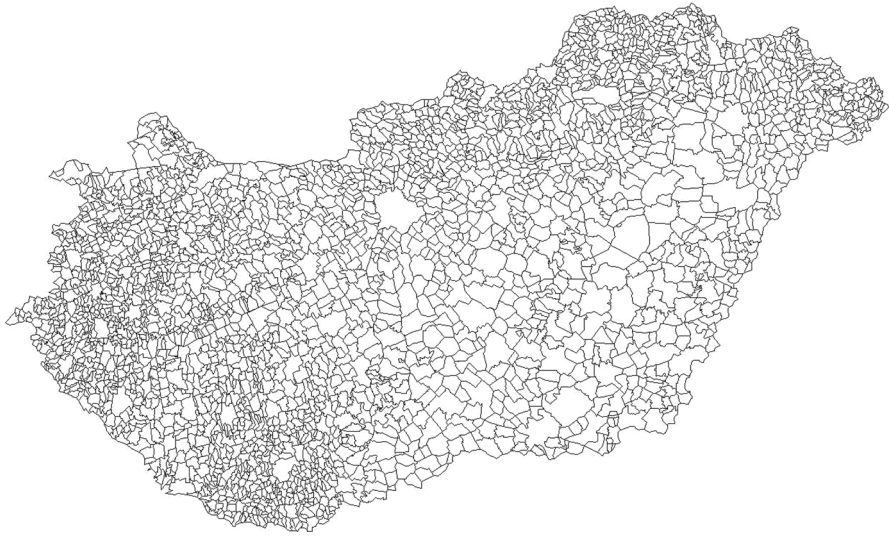


Fig. 1 Map of Hungary at municipal level

heterogeneity of the size of areal units, such as municipalities and regions. We will discuss similarities and differences with regard to other methods, in particular, with respect to [Briant et al. \(2010\)](#) whose method we replicated.

In what follows, we first describe key issues that shed light on why measurement methodology matters. Section 3 introduces the methodology we developed, and in Sect. 4, we illustrate the method introducing two examples and present the empirical results and compare our method with some existing approaches. The last section concludes.

2 Considerations for measurement

Before discussing the method we are proposing, let us present some key considerations that motivated the procedure. First, we argue that unit size matters and there is a need to compare results across spatial units of different sizes. Second, uneven and non-random distribution of unit size matters as well. Third, in the presence of policy aspects, we will be able to understand the influence of administrative borders when investigating political-administrative unit size effects. Finally, instead of a randomly chosen robustness test, we should be able to perform statistical inferences and hypothesis testing. At each subsection, we will present some arguments why a particular measurement issue may matter.

2.1 Unit size matters, scalability is key

There is substantial evidence suggesting that the impact of spatial externalities decays rapidly with distance, and measured elasticities will be altered based on the unit size

of measurement. Hence, the presence of externalities may be detected at some level of spatial unit, but not at another. This issue is partly an econometric problem: a mismatch between the spatial unit of observation and the spatial extent of the phenomena under consideration will result in spatial measurement errors and spatial autocorrelation between these error terms in adjoining locations (Anselin and Bera 1998). Uncertainty and practical differences regarding unit size may hinder comparing evidence across countries or regions.

Consider the example of EU regions so that we can see how the level of aggregation changes correlation inferences as well. Figure 11 shows the population density distribution of Western European countries at NUTS1 and NUTS3 level (see Fig. 11 in “Appendix” section). One can see that even within the same administrative level the size and shape of the basic units differ a lot. For example, the correlation between population density and GDP per capita across European regions at NUTS1 level is 0.74. The same relationship at the more disaggregated level of NUTS3 administrative units is much weaker, i.e., 0.54.⁴ This difference may be of purely random nature or related to the spatial decay of some relationships. For instance, the impact of an externality may weaken over distance for a wide range of activities such as the trade of intermediate inputs, productivity and wages and knowledge spillovers.⁵

Hence, the first important requirement is for a methodology that can be scaled by size of units of observation (i.e., compare distributions from two randomization runs (at $L_1 \times L_1$ -sized grid and another one at an $L_2 \times L_2$ grid), thereby allowing for a direct comparison.

2.2 Uneven and non-random distribution of unit size matters

The second issue is that spatial units are not only of uneven size, but they are non-randomly located, potentially biasing results. The spatial correlation of unit size (such as very small villages are collated in a particular geographic region) may stem from several sources: maybe first geography (lakes or mountains) prompts people to organize themselves into small units, or dense urban areas set up small but populous spatial units. The political use of spatial units may also lead to some clustering that is possibly related to historical population density or institutions such as fairs. Districts tend to have an equalizing role across population. In some countries and at some levels of aggregation this could be a very minor issue. In France, for example, departments are fairly equal sized.⁶ However, this is not the case in most countries and regions. In fact,

⁴ We use the 2007 euro per capita measure of GDP and the 2006–2008 average population density data from the Eurostat regional databases. The countries that constitute our sample: the UK, Germany, Belgium, France, Spain, Portugal, the Netherlands and Italy.

⁵ For instance, trade proximity matters for localization for about 50km as suggested by Duranton and Overman (2008) for manufacturing industries in Britain. Rosenthal and Strange (2008) estimated that in the USA wage increasing effect of being close to educated people falls to just 25% as the distance rises from 5 to 15 miles in the USA. Lychagin et al. (2010) found that the crucial range for proximity to laboratories is limited to 100–200km. Andersson et al. (2016) document and attenuation of wage density elasticity even within cities.

⁶ Note that, shape of units may be of interest. Briant et al. (2010) showed it does not matter. However, even though shape is not of our immediate concern here, the method presented caters for different shapes as well.

when looking at the EU NUTS3 regions above and observing the representation of population density distribution, it is obvious that the dense regions are much smaller. Importantly, not only do we see that small and large regions are clustered (i.e., North European regions are small, and Spanish regions are large), but that this is correlated with first geography (Spanish mountains, access to Atlantic trade routes) and urbanization (abundance of mid-sized North German cities, possibly related to the system of *Länder*).

Take another example. In this paper, we will work with data from Hungary, where the municipal and micro-regional structure is rather unbalanced (See Fig. 1 in Sect. 4.1)—unlike in France, where the large administrative regions called *départements* are of about equal size. We see an average municipal size of 15–20 km² in the southwest and 70–75 km² in the northeast, and there is a great deal of variation in the size of micro-regions, as shown in Table 3. The small size of typical administrative units in the southwest is correlated with rugged terrain, history, and agricultural traditions. These are all issues that may bias the estimated coefficients.⁷

Therefore, the more disaggregated data one uses, the more the number of dense regions increases in the sample. This may affect inferences about concentration and also gives less weight to less populated areas, and given its correlation with other variables such as first nature geography, it may bias results. In other words the size distribution of density and hence productivity (or wage) are endogenously determined by the aggregation level of administrative units. The more densely populated a location is at the most disaggregated level (municipality), the most likely it is to be aggregated up with other locations at the higher aggregation level (region).

Consequently, the second expectation is the ability to treat the clustering of large or small units and thus the ability to compare aggregated units of the same size rather than the same sub-units.

2.3 The role of administrative borders can be directly investigated

The third issue is related to administrative unit borders.⁸ Regions are often units with political or administrative roles, where local bureaucracies set policies on the labor market (often by regional employment agencies), taxes (such as Bacher and Brühlhart 2013), firm subsidies (by regional development agencies) or regulation (e.g., by regional environment protection agencies). Regional differences may play an explicit role when grants are given based on some average regional development measure such

⁷ When regressing the (log) number of municipalities within a district on (log) wages, we find a negative correlation. This is primarily due to the fact that there are plenty of small municipalities close to Vienna with easy market access and higher wages. Using a cross section, we have: $\log(\text{Num}) = 11.8 + 0.8 \times \log(\text{Wage})$, significant at 5%; $\log(\text{No}) = 1.41 + 0.05 \times \log(\text{Wage}) + 0.45 \times \log(\text{Distance}_{\text{Budapest}}) - 0.24 \times \log(\text{Distance}_{\text{Vienna}})$, where both distance variables are significant at 1%.

⁸ In some countries, like Hungary, municipality also contains the agricultural land. In Hungary, Eastern municipalities have larger plots of land and hence, Western municipalities will be calculated as denser.

as per capita GDP, income or unemployment. It may be an explicit research question to seek to understand the effectiveness of such policies.⁹

Hence, the ability to compare administrative regions with a set of artificially defined spatial units matching the average regional size may offer a useful insight. As a result, the third requirement is hence the capacity to compare administrative regional units with artificial units where the size of the units are kept to match (on average) the administrative regional size. With our model, we can compare this distribution of coefficients gathered from the randomized $L \times L$ grid to the single measure, say, at the administrative regional unit level. This allows us to understand to what extent we have a special realization of results; whether can we claim, for instance, that 95% of the grids yield a smaller or larger coefficient than the administrative one.

2.4 Statistically differentiating estimations

A key drawback of estimations using administrative units is the lack of robustness. One estimates a model and gets a point estimate that may be influenced by policies, some special first geography issue (rivers, mountains). The argument that location specifics may influence estimation is indeed a core idea of the Modifiable Unit Area Problem or the MAUP. As a result, any particular aggregation may suffer from a set of idiosyncratic shocks, for example, particularities of regional border. To see the robustness of a particular relationship, one may need to go beyond comparisons and understand the statistical significance of the differences in elasticities measured at various levels of aggregation. To allow for comparisons across different sized units as well as artificial versus administrative units, the method will have to be able to generate randomized aggregation outcomes allowing to create standard errors for regression outcomes at various levels of aggregation.

3 The “grid and shake” method

This section presents a new method, that we dubbed “grid and shake.” Before the actual estimations, several steps are needed. First, we digitize the map by placing markers evenly on the country and hence pinning down municipalities. Second, we define a grid made up of $L \times L$ squares and place it over the map. Our markers will help assigning each municipality to a particular unit in the grid. Third, we define the grid unit size based on some economic consideration, such as matching the average size of administrative units. Fourth, we shake this grid and thus, randomly create possible realizations of spatial units. Finally, we make cosmetic adjustments around the country’s zigzagging borders to avoid having too small units. When running regressions, we can repeat any given regression on all realizations and thus get a distribution of elasticities. In what follows, we present step-by-step details using Hungary as an example to illustrate the method.

⁹ See Dewhurst and McCann (2007) looking at a relationship between size and specialization in different types of administrative levels in the UK, or Devereux et al. (2007) studying UK “Regional Selective Assistance” grants given to regional political units.

3.1 Setting up the grid

To collect municipalities into spatial groups of an approximately equal size, we use an idea analogous to offset printing technology. First, we place markers arranged in a raster-like manner on the map with the lowest available units such as municipalities or ZIP codes. Here we use the municipality-level map of Hungary.

Markers are defined on the positive quadrant of a coordinate system¹⁰ where the origin is defined as the (0, 0) km. They are placed rather close to each other (800 m) such that the smallest municipalities have at least three points within the boundary of the polygon that define them. In turn, the markers (we have 153,113 marker points for the 93,030 km² of Hungary) uniquely identify the spatial territory of a municipality. The markers are matched to the polygon of the municipality.¹¹

As markers are assigned to municipalities, they make it possible to group municipalities into larger spatial units by aggregating them. If we place a grid consisting of size L squares over the map, we can identify the markers that belong in the same square. The square of the grid is defined by a number pair (g_x, g_y) . We assign each marker to a grid square with: $g_x = \text{int}(x/L) + 1$ and $g_y = \text{int}(y/L) + 1$, where x and y are the coordinates of the markers, L is the length of the side of the square, and int is a function that takes the integer part of a number. With the coordinates of the markers being given in km from the origin (0, 0), g_x will then give the number of grid square we need to put on the map starting from zero to arrive at the longitude of the marker (x, y) . That is, if the marker coordinates are, say, 23 and 42, then using a grid size 10, the marker will be in the 3rd grid square from the left and the 5th from the bottom.

As a next step, we can form a new spatial unit by joining municipalities whose points are in the same grid. If the points that define a municipality belong to several grids, then the municipality joins the spatial unit that holds most of its points (i.e., the largest territory). This method gives an alternative realization of larger spatial units anchored by the starting point (0, 0).

Importantly, the size- L grid could be placed on the map at random locations, yielding different versions of grid. This feature will be used later to generate a set of grids allowing for drawing a distribution of regression model estimates.

3.2 Defining grid sizes

To analyze the role of size and administrative units, we create grids in three sizes: small, medium and large. Small grids are 15 km $\times$ 15 km, and it takes roughly 440 of such squares to cover the country. This is the size that fits the top 1% of municipalities (mostly large cities such as Szeged, Nyíregyháza or Miskolc). By aggregating small villages and keeping large cities as they are, a realization with grid size of 15 km

¹⁰ Mercator WGS84 EPSG:41001 projection.

¹¹ Polygon matching can be done in a GIS mapping software; we used Mapinfo. In Fig. 9 in appendix, they are illustrated by using a small part of southwest Hungary close to the Croatian border.

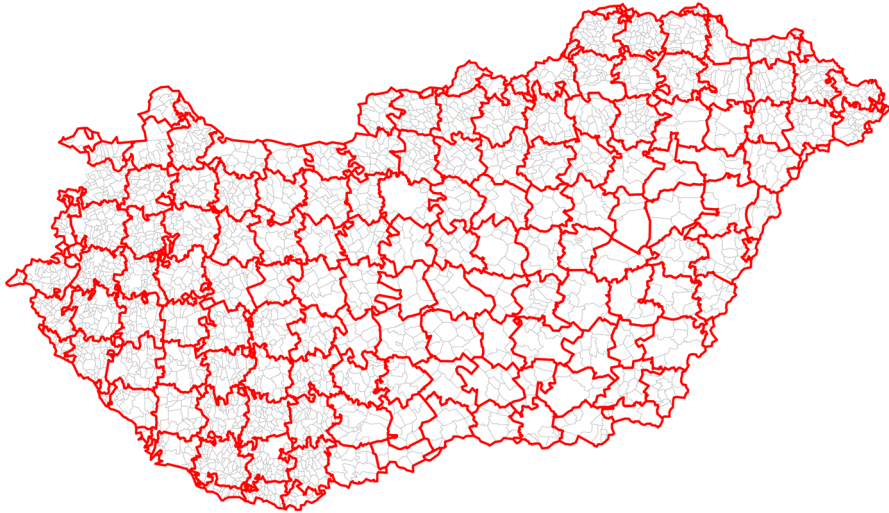


Fig. 2 A realization of 26×26 km units

produces a spatial division simulating a case when municipalities were the same size as large cities.¹²

The medium-sized grid is made up of 26×26 km². We chose this size to match the number of micro-regions, i.e., 150. This is apparent from Table 3 which summarizes the area and count data for different spatial aggregations. The largest size of grid squares is 39×39 to match the largest (the average of top 5) micro-regions. This will allow us to simulate what would happen if all micro-regions were equally sized and as large as the largest ones Hungary has. Figure 2 illustrates a realization of the grid based on 26×26 km sized spatial units.

To have a better sense of how aggregation works, we took one grid and calculated the number of municipalities per grid. Figure 3 shows the distribution of the number of municipalities for both NUTS4 micro-region level and the artificial grid. As we argued earlier, Hungarian municipality sizes vary a great deal, moreover, non-randomly, hence the deviation. About 30% of the units contain only a few number of units (no more than 10), while about 25% include many (31+). Note that as the graph suggests, the distribution of the number of municipalities per unit (both when considering the first realization and the average 300 permutations) matched fairly closely that of NUTS4 regions.

Having these three artificial grids of sizes, with 15×15 km, 26×26 km, 39×39 km squares, we can now easily compare the spatial features of relationships, such as the spatial magnitude of agglomeration spillovers.

¹² This approach would in some way, mimic the Swedish example, where 290 municipalities have been defined with a minimum population requirement, yielding similarly sized areas—except for the large Nordic regions.

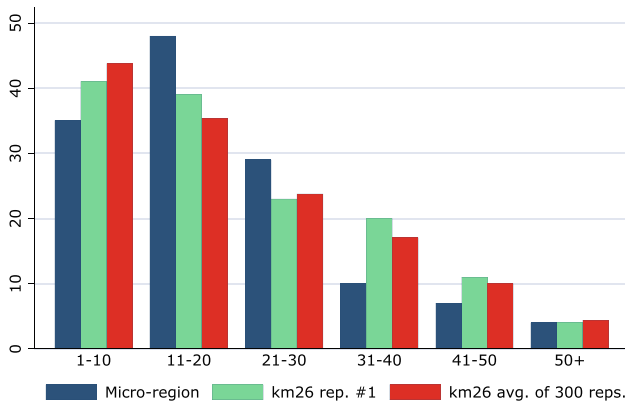


Fig. 3 Distribution of units that include few and many municipalities

3.3 Border correction

A disadvantage of the method described above is that small areas are likely to get drawn by the border. To partially offset this, we considered all areas (created by the grid) that are smaller than a pre-determined threshold (set as 10% of $L \times L$ km) and added it to a randomly chosen neighboring region. This border issue is more important for small countries/larger grid units and with relatively more zigzagging or just non-horizontal/vertical border line features.¹³

3.4 Randomization

The most problematic aspect of both any given administrative regional setting and the artificial grid we propose is the arbitrary boundaries of units. When considering the robustness or validity of the results at a certain level aggregation or finding out about the importance of administrative units, the choice of spatial units may be important. As we will show later, two randomly placed grids may often yield rather different outcomes. To treat this arbitrary nature, we will randomly generate 300 possible grids—by first placing a grid of $L \times L$ km sized units and then “shaking it” over the map to yield different sets of aggregations.

The procedure works as follows. We select a random starting point by adding independently drawn random numbers ξ and ζ to the (0, 0) starting point which are uniformly distributed over the 0 to L interval. Then, using the algorithm described in Sect. 3.2, we assign markers to $L \times L$ km sided grid square. The bundle of municipalities included in any new grid unit will be different from the one defined by the

¹³ This is illustrated in the left panel of Fig. 10 in “Appendix” section, where the random grid joins only three municipalities at the border, including Órtilos, Zákány, Surd and Mezőkeresztúr. The right panel of Fig. 10 shows the adjusted spatial units. In this case, the four municipalities are added to the larger region to the east, hence increasing its size.

starting point (0, 0). It is worth noting that picking random numbers from (0,L) is enough as $\xi + L$ and $\zeta + L$ would yield the same realization as ξ and ζ .

In the first step of the procedure, we assign a grid (a pair of numbers) to each point so that we get each municipality to belong to a specific unit in the grid. We carry out border corrections when necessary. In the second step, the randomized grid placement is repeated a thousand times. Third, we open a spatial structure, calculate several variables of interest and run regressions and save the outcome. Fourth, we collect estimated coefficients, generate the distributions and draw conclusions. Finally, we take these saved coefficients and consider the distribution at various grid sizes, with $L = 15 \text{ km}, 26 \text{ km}, 39 \text{ km}$.

4 Empirical applications and discussion of results

In this section, we present two empirical applications and show the procedure in action. We present the details of the Hungarian data we employ and introduce two topics that use regionally aggregated data. We start by looking at how wages are correlated with density followed by looking at how agglomeration may be correlated with bolder entrepreneurship measured by new firm formation. For both of these cases, we use the artificial grid method to understand differences in various measures of agglomeration. This is followed by a discussion of external validity of our observations and finally, a comparison of insights with lessons from existing methods.

4.1 Data

We use Hungarian firm-level balance sheet data from the Central Statistical Office to obtain firm-level information and resort to information from 1999 for the cross section OLS. Firms include all double accounting manufacturing companies with at least five employees. We have the same information for 1998 as well, allowing us to create variables for new firm formation. Firm-level financial data are matched with location information at municipal level.¹⁴ Firm-level data are then matched with municipality-level information on municipality size and population from the T-STAR database.

Wages are calculated from balance sheet data using additional information on full time employees and total wage cost. The proxy for human capital is obtained from the Hungarian Labor Force Survey (HLFS). The survey contains a sample of workers in the manufacturing industries and documents various employee characteristics. The information includes their highest educational attainment, which is used here as a proxy for human capital. The data connect workers to municipalities; however, their actual employers are not identified, which prevents linking the firm-level information with worker data. Using the HLFS sample, the municipality-level average of the highest grade of schooling variable can be calculated to serve as the proxy for the quality of

¹⁴ Unfortunately, we only know about headquarters. However, this is not a major problem for manufacturing firms in Hungary: in their appendix, Békés and Harasztosi (2013) showed that only 7% of firms have multiple sites and that, on average, these firms have 1.15 plants. This may give rise to a relatively small potential bias only.

the labor force. This human capital proxy shows a considerable variation from 7 to 14 years of education attained.¹⁵

4.2 Two models: wage and firm formation elasticity of agglomeration

One key evidence in the economic geography literature is that firms operating in more densely populated areas show higher productivity and also pay higher wages. The positive correlation between density and productivity stems from both natural advantages and externalities that boost productivity as was first described by Marshall (1920). The Marshallian externalities play an important role in the micro-founded explanations for growth, innovation and trade theory, and understanding their strength is important in order to evaluate economic policies that, for example, aim at cluster formation or at helping backward EU regions to close the gap with more developed regions. Typically, the literature finds that, on average, productivity of firms located in a region with 10% larger population tends to be 0.4–1.2% higher, and this higher productivity will then be equally translated into higher local wages. For a thorough meta-analysis on results, see Melo et al. (2009).

In our exercise, we opt for considering wage as the dependent variable, as this is the most widely investigated relationship and hence allows for international comparisons. We estimated the agglomeration effect on the average municipality wage:

$$Wage_i = \beta_1 DENSITY_i + \beta_2 EDUC_i + v_i \quad (1)$$

where i is the index of the spatial unit, $Wage$ is the average wage of employees, $DENSITY$ is the density of working age population in the municipality, and $EDUC$ is a proxy for education levels. This is estimated with OLS on a cross section of 1999 data, with all variables used in logs.

The results are presented in Table 1; the first two columns show regression results on actual administrative (municipal and micro-regional) units, and columns (3) to (5) show results on the artificial grids of 15×15 , 26×26 and the 39×39 grid, respectively. For all artificial grids we display the mean of estimated standard errors followed by the standard deviation of the estimated coefficient.

The table suggest that at all estimation levels, there is a positive correlation between the average wage in an area and its density (size of population holding area size constant). The estimated coefficient is lowest when using municipal-level data (5.5%), followed by small artificial units (7.7%), while the estimate for micro-region and medium-sized artificial units (11.8%, 12.7%) is about double the size of the municipal-level estimate. Having artificial units allows us to compare these figures attaching statistical significance to any gaps we may find, and we will discuss the comparisons in Sect. 5.

¹⁵ Note that the variables are recalculated on different spatial aggregation levels and artificial spatial units by recalculation and using frequency weights. For example, if the average year of schooling (ys) in municipality i is ys_i calculated over n_i individuals, then the average year of schooling for spatial unit that includes municipalities i and j will be $(ys_i \times n_i + ys_j \times n_j) / (n_i + n_j)$.

Table 1 Wage regressions

Administrative units			Artificial grid units			
Units	(1) Municipal	(2) Micro-region		(3) km15	(4) km26	(5) km39
Density	0.055	0.128	Density	0.082	0.116	0.137
SE	0.007	0.021	(mean) SE	0.011	0.019	0.026
			SD coeff.	0.006	0.008	0.015
School yrs.	0.966	0.820	School yrs.	1.098	1.123	0.979
SE	0.041	0.355	(mean) SE	0.169	0.442	0.664
			SD coeff.	0.121	0.219	0.510
			Replication	300	300	300
Obs.	2872	149	Obs	437	150	73
R-squared	0.244	0.383	R-squared	0.240	0.361	0.393

All variables are in logs. Density is the size of local working age population divided by the area in km². Years of schooling is average of local population. Robust standard errors are displayed. In grid models we report two statistics for standard errors: (mean) SE is the mean of robust standard error of the coefficient over 300 replications, while the SD coefficient provides the statistic for the standard deviation of the coefficient itself calculated over 300 replications

Note that in municipality-level regression the number of observations is 2873. This is smaller than the total number of municipalities (3125) as there are municipalities without any identifiable manufacturing activity. This is an important reason behind the use of some aggregation technique, especially in a highly fragmented country with unequally sized units.¹⁶

As our second topic, we will consider new firm formation (or entrepreneurship) as an example of a process with great spatial variation. Regional variation in new firm formation has been amply studied. Key explanatory variables include measures of unemployment, population density, industrial restructuring, and availability of financing ([Armington and Ács 2002](#)), income as well as market dynamics such as exits and change in industrial structure ([Sutaria and Hicks 2004](#)). Infrastructure, such as motorways, may also affect firm entry and location ([Holl 2004](#)). In a close exercise to ours, [Amrhein and Flowerdew \(1992\)](#) use a Poisson model to analyze labor movement. Since policy interventions enhancing entrepreneurship abound at local as well as regional levels, the evidence from such exercises may have policy implications as they can answer questions about the usefulness of regional policies and grants offered by regional authorities. We can test whether regional boundaries matter by investigating whether the policy impact is bounded by administrative unit lines.

¹⁶ This selection issue might affect the comparison between municipalities and 15 km grids because the latter will include the areas of zero employment areas in the denominator of the density measure, but it does not affect grids of larger sizes, like the 26 and 36 km grids. The sign of the bias caused by the selection is ambiguous and is an issue in the western part of Hungary, owing to the highly fragmented municipality structure.

Table 2 Entry regressions

Administrative units			Artificial grid units			
Units	(1) Municipal	(2) Micro-region		(3) 15 km	(4) 26 km	(5) 39 km
Density	1.377	0.917	Density	1.321	1.257	1.146
SE	0.079	0.172	(mean) SE	0.074	0.096	0.219
			SD coeff.	0.027	0.039	0.085
Wage	0.608	0.554	Wage	0.526	0.392	1.664
SE	0.200	0.716	(mean) SE	0.266	0.392	0.654
			SD coeff.	0.110	0.152	0.358
			Replication	300	300	300
Obs.	2872	149	Obs	437	150	73
R-squared	0.467	0.379	Ps. R-squared	0.605	0.615	0.614

All variables are in logs. Density is the size of local working age population divided by the area in km². Years of schooling is average of local population. Robust standard errors are displayed. In grid models we report two statistics for standard errors: (mean) SE is the mean of robust standard error of the coefficient over 300 replications, while the SD coefficient provides the statistic for the standard deviation of the coefficient itself calculated over 300 replications

Here we estimate a Poisson model, where we regress the number of new firms on population density and labor market conditions proxied by wages. Our variable in focus is once again density captured by the population of the area.

$$Pr(y_i = N) = f(\lambda_i)$$

$$\lambda_i = \beta_1 DENSITY_i + \beta_2 Wage_i \quad (2)$$

Results are presented in Table 2, with columns (1) and (2) reporting results on administrative and columns (3)–(5) on artificial units. Estimated coefficients of the population coefficient are greater than unity, suggesting a net positive relationship between population and firm formation.

4.3 Grid and shake in action

In this subsection our aim is show what kind of statements are possible with the use of the grid and shake method. To do this, we will rely on estimated agglomeration elasticities from the two exercises presented earlier. Estimates are shown regarding two types of administrative units (municipalities and micro-regions) and three artificial grids, made up of 15×15 km, 26×26 km and 39×39 km units. For all artificial grids, we generated 300 replications—regressions were repeated 300 times, yielding a distribution of elasticities. To demonstrate our points, we devised four graphs with distributions of estimated coefficients. Figures 4 and 5 depict elasticities for the first example on density and wage, while Figs. 6 and 7 show them for the second one with firm entry and density.

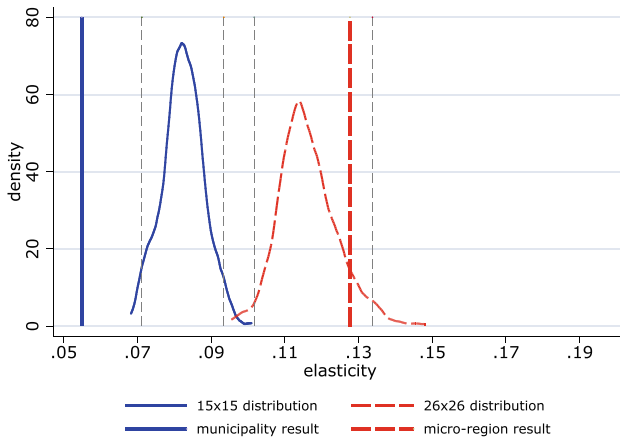


Fig. 4 Agglomeration externality—administrative versus artificial units. This figure shows the coefficient estimate on population in Model 1 (agglomeration elasticity of wage). The thick blue (solid) vertical line gives the point estimate for the municipality estimate. The thin blue (solid) line gives the distribution of point estimates over 300 point estimates on the 15×15 grid. The thick red (dash) vertical line gives the point estimate for the micro-region estimate. The thin red (dash) line gives the distribution of point estimates over 300 point estimates on the 26×26 grid. The very thin dashed lines are 5% confidentiality thresholds

In what follows, we discuss three applications: (i) the role of fragmentation comparing municipalities with a grid simulating average city sized units (15×15 km); (ii) looking at the impact of the size of units, comparing three grid sizes; and (iii) the role of administration by comparing micro-regions with artificial units sized 26×26 km that match the average size of micro-regions.

4.3.1 Fragmentation of municipalities

Hungary has a rather fragmented municipal system with 3125 local units for 10 million inhabitants. For the similarly populous (but much larger) Sweden, we find 290 municipalities, Portugal has 309, while Austria has 2300—the highest number for medium-sized countries.

The first application is hence related to a lower level of public administration, as we investigate how this fragmentation may affect the estimated relationship between agglomeration and wages as well as new firm formation. To demonstrate this we can now compare regression results at the municipal level with estimated elasticities from regressions on the 15×15 km grid. For instance, such a set of results would allow comparisons with model estimates regarding other countries with less fragmented systems.

In this dataset, fragmentation of the local political-administrative unit implies that locally estimated agglomeration elasticities are different to what they would be in the case of artificially aggregating small units, namely creating (on average) 440 artificial units instead of actual municipalities.

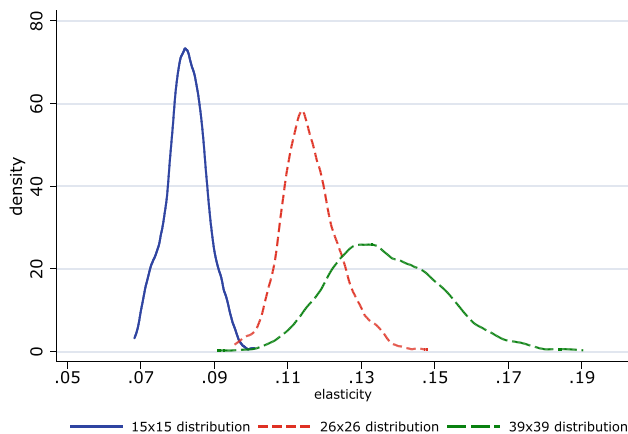


Fig. 5 Agglomeration elasticities at various grid sizes. This figure shows the coefficient estimate on population in Model 1 (agglomeration elasticity of wage) over various grid sizes. The blue (solid) line gives the distribution of the coefficient over 300 estimations on the grid size of 15×15 km. The red (short dash) line gives the distribution of the coefficient over 300 estimations on the grid size 26×26 km. The green (long dash) line gives the distribution of the coefficient over 300 estimations on the grid size of 39×39 km

For both the wage OLS and the firm entry Poisson regressions, we find that the estimated elasticity for the municipal level is lower than the 95% of elasticities calculated using artificial units, see blue (solid) lines in both Figs. 4 and 6. The difference is particularly strong for the wage regressions: the estimated elasticity is 5.2 at the municipal level and is lower than the 95% lower bound of 6.6 of the 15×15 km sized artificial unit level.

Importantly, this means that, purely for methodological reasons, estimations carried out at the same nomenclature (NUTS5—municipalities) in countries with larger administrative units (like Sweden or Portugal) are likely to be above the one we obtained for Hungary (or those that may be estimated in Austria).

4.3.2 The size of spatial units

The second issue is related to the size of units of observations by comparing results from grids of 15×15 km, 26×26 km and the 39×39 km sized units. Figure 5 shows the population elasticities at various levels of grid size for our example 1 (wages) and Fig. 7 plots results obtained from example 2 (firm entry).

As for wages, we find that the population elasticity is different at 15×15 km and 26×26 km size, but not beyond. (Note that as suggested by Table 1, the relationship with years of schooling varies between 26×26 km and 39×39 km unit sizes.) Regarding firm entry, point estimates suggest that elasticities are different for the 39×39 km grid from the other two, but given the larger variation of estimates, these are not statistically different. The possibility of establishing such arguments is the main reason behind our randomization method (However, as shown in Table 2, the

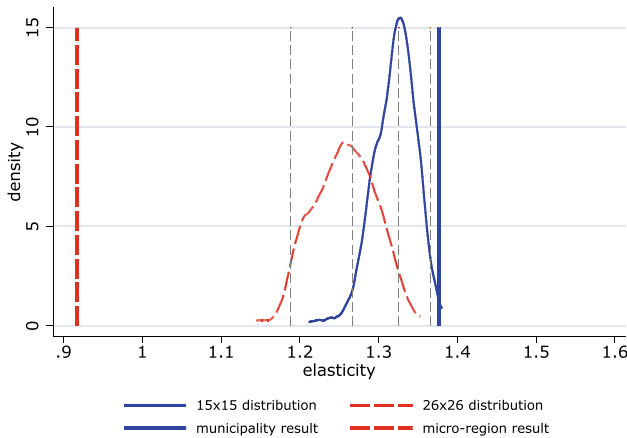


Fig. 6 Agglomeration externality—administrative versus artificial units. This figure shows the coefficient estimate on population in Model 2 (agglomeration elasticity of firm formation). The thick blue (solid) line gives the point estimate for the municipality estimate. The thin blue (solid) vertical line gives the distribution of point estimates over 300 point estimates on the 15×15 grid. The thick red (dash) vertical line gives the point estimate for the micro-region estimate. The thin red (dash) line gives the distribution of point estimates over 300 point estimates on the 26×26 grid. The very thin dashed lines represent the 5% confidentiality thresholds

elasticity of wage level on firm entry on the large grids is statistically different). These results broadly conform with estimates found earlier.¹⁷

4.3.3 The role of political-administrative division

As final application, we now consider the role of administrative units by comparing elasticities measured at the micro-region level with a grid of 26×26 km sized units. Table 1 shows that for model 1 (wage regressed on population), we find an estimated coefficient of 12.7% for the micro-regions, and this may be compared with the *mean* estimated coefficient of 11.8%—averaged out on the 300 replication of the 26 km grid. Figure 4 depicts this with dashed lines, suggesting that while there may be a small difference, the estimated coefficient of the administrative unit is not statistically significant as compared to the artificial set of results. Hence, we cannot reject the null of zero effect of administrative units of micro-regions compared to the artificial units of equal average size (26×26 km).

The story is somewhat different when we consider firm entry (Fig. 6). All estimated coefficients are significantly greater than unity. However, the equality of estimated coefficients is here rejected at 5% confidence level, suggesting that there is a small, but statistically significant difference between elasticities at artificial units compared

¹⁷ Comparing them with the results presented by Briant et al. (2010) using a different aggregation methodology, we find that the order of elasticities is different. In this paper, we find coefficients rising with size, but a significant difference can only be established between small- and medium-sized units. Briant et al. (2010) found no linear relations, and the difference was only between their medium- and large-sized units.

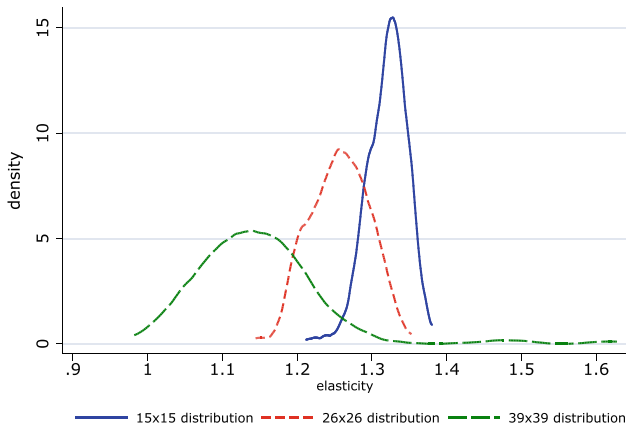


Fig. 7 Agglomeration elasticities at various grid sizes. This figure shows the coefficient estimate on population in Model 2 (agglomeration elasticity of firm formation) over various grid sizes. The blue (solid) line gives the distribution of coefficient over 300 estimations on the grid size 15×15 km. The red (short dash) line gives the distribution of coefficient over 300 estimations on the grid size 26×26 km. The green (long dash) line gives the distribution of coefficient over 300 estimations on the grid size 39×39 km

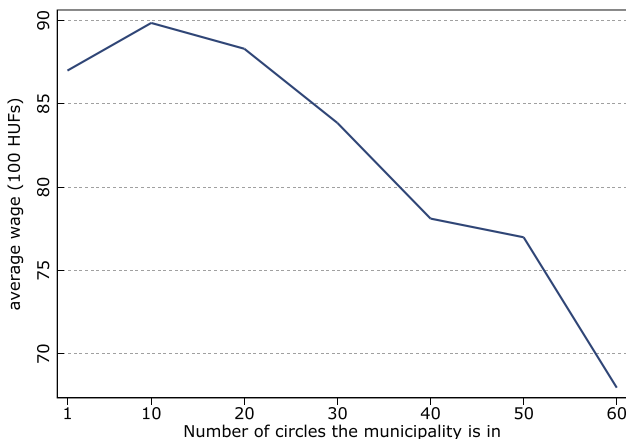


Fig. 8 Frequency of a municipality being included in a circle and its mean wage. The number of circles is created by first calculating for each municipality the number of circles this municipality is included in, and then grouping them by this count. Average wage is calculated for groups of municipalities based on this count as well

to administrative regions. Note that estimates at different artificial units vary more than earlier, and tails are longer, too, as depicted in Fig. 7.

This suggests that finding statistical difference depends on the nature of the exercise. In our case, we found that elasticity of population density is lower at the administrative unit—compared to similarly sized artificial units. Thus, it is possible that some policy might have been in play here.

4.4 Generalization—outside validity

To generalize our results, we test the null hypothesis that administrative borders do not matter in countries represented by our Hungarian data. Viewed this way, the coefficient estimates from our data, including the estimates from each of the 300 Grid and Shake permutations, are just one realization from their respective distributions. The entire distribution of potential point estimates under this null hypothesis is therefore an all-encompassing distribution of the coefficient estimates across all permutations. The question is then the extent to which the distribution of the coefficient estimates with the true administrative boundaries overlaps with this all-encompassing distribution. An intuitive comparison suggests that we can reject the null hypothesis if the two distributions are different from each other.

To estimate the encompassing distribution across all permutations we took a numerical approach. In each permutation we took 50 random points from the coefficient distribution implied by the point estimate and the standard error, assuming a normal distribution. The all-encompassing distribution is then estimated by the empirical distribution of these 300×50 coefficients. The distribution implied by the administrative boundaries is simulated numerically in the same fashion, with 50 points drawn randomly from the normal distribution around the point estimate with the standard error estimate. The original and this extended distribution are shown in Fig. 12 in “Appendix” section. Looking at this graph as well as results of the two-sided Kolmogorov–Smirnov test, a systematic test of the equality of the two distributions, we now cannot reject the null hypothesis.¹⁸

In addition to comparing a point estimate of the administrative unit to the distribution of estimates of an equally sized artificial grid, this procedure also allows for a comparison of distributions by two differently sized units. To illustrate both these approaches, we created a chart denoting means and 90 and 95% confidence intervals for both administrative classifications and simulated distributions with a single 300 replications (as in Fig. 4) and 50×300 replications as in Fig. 12. The graphs 13 hence shows six lines, each with symbols denoting confidence intervals. In addition to the previous observation, we see that administrative versus artificial grid estimates now cannot be told apart not only for the micro-region but also for the municipalities. Furthermore, now the difference between the estimates from the two artificial grids is not significantly different from zero. This suggests that our particular finding on the difference between differently sized regions is just indicative for other countries.

4.5 Comparison with other methods

Finally, let us compare this method with some existing methods. In particular, we will discuss issues with aggregating units by drawing a circle around observations and discuss differences with respect to the method proposed by Briant et al. (2010).

¹⁸ KS results are available on request.

The most basic method is aggregating units (markets) within a circle with a given radius around an observation. For instance, [Martin-Barroso et al. \(2010\)](#) used this method to correct for the agricultural area of municipality size when estimating agglomeration elasticities. This simple spatial aggregator function method assumes that we are uncertain about the market size or the extension of some externality and consider access to be symmetric around the observation. In this process, we set up circles around each observation and aggregate units. The advantage of this method is simplicity and flexibility regarding spatial unit size.

The disadvantage is a bias introduced by the oversampling of areas composed of many small municipalities. To better see this, we look at how the circles method would work in Hungary. There are 3125 municipalities, so we create 3125 circles with a center located in each of these municipalities. The circles have a radius of 14 km so that its area would match the average size of NUTS4 regions as well as the average area of the size of our grid of 26×26 km. For each municipality, we calculate the average wage, as well as count the number of circles the municipality is included in. In the case of Hungary, we find a correlation: richer municipalities tend to be counted less frequently on average than poorer ones, thus biasing estimates, see Fig. 8. This does not happen when estimates are based on disjoint areas, where all municipalities are counted only once.

Spatial dynamic or distributed lags models have also been used to capture the potential correlation of the independent variable across space. For instance, [Cainelli et al. \(2014\)](#) analyzed firm exits as a function of agglomeration and used neighboring Italian regions in a dynamic panel model estimated by GMM. Similarly to our model, this allows the testing of decay over space. However, the number of neighbors will be determined independently of the size of adjacent units.

Our approach of creating spatial districts is closest to the “dots-to-boxes” (DtB) procedure developed by [Briant et al. \(2010\)](#), who investigate the effects of MAUP on inferences using French administrative units, and applying two approaches. First, they place a grid over France which creates the same number (N) of new spatial units as the one they wish to compare it to. This creates square-shaped units with only one realization. Second, to match the numbers of existing administrative units, they randomly pick basic units of the same number (N) and merge them with neighboring units. This procedure results in differently shaped and sized N spatial units depending on which seeds are chosen and on the sequence how neighboring units are added. The size also depends on the spatial distribution of the size of the basic units. A similar method to the DtB approach is used by [Menon \(2012\)](#), who also creates artificial spatial units and does a simulation exercise keeping the size distribution of the administrative units fixed. While he aims at recreating a size distribution, we simulate how elasticities would change should unit size change—while keeping units equally sized in one go.

The size distribution of the basic units will influence the size of the created spatial units. In France, due to the roughly equal size of basic spatial units, this is not a problem. However, the same method would not be entirely satisfactory in the case of Hungary or for NUTS regions in the European Union. Consequently, our method is an alternative to that of [Briant et al. \(2010\)](#) and can be applied to countries or federations with basic spatial units that are rather heterogeneous in size and still can

test robustness. To this end, we repeated our core exercise (Fig. 4) now applying the method used in Briant et al. (2010). The lower panel of Fig. 14 in “Appendix” section shows a map with artificial units this time created by the DtB algorithm. Seeds were fixed to generate 150 units to match our core grid of artificial units depicted in the upper panel of the same figure. There are two key observations to make when comparing these two graphs. First, while in some areas the units are similar (North–East), in many instances, there is a great deal of variation in size and form. Second, as expected, DtB will generate small (south–west) and large (center–east) units owing to a large variation in Hungarian municipality unit size. For instance, our model generated 93 units west of the Danube, while DtB generated 57 units.¹⁹

To gain further insight, we reproduced Fig. 15 now also depicting the distribution of coefficients by the Briant et al. (2010) method. We can see that the median point estimate is somewhat lower, and more importantly, the standard deviation is much higher (0.7 vs 1.4%). Both of these features may be the consequence of unequally aggregated spatial units (municipalities). For the same figure, we also reproduced results using the Circle method, using a circle with a radius of 14 km (to match the 26×26 km artificial grid size). Unlike for the DtB, this generates a single coefficient estimate that can be compared to the distribution. It is smaller than the micro-region point estimate, but both are in confidence interval of our core estimate.

5 Conclusions and practical applications

In this paper, we proposed a flexible, transparent, simple and relatively fast²⁰ method to test the robustness of empirical investigations at various levels of spatially aggregated data. In its ability to generate a distribution of estimated parameters (such as agglomeration elasticities), it is an alternative to Briant et al. (2010), while in terms of taking distance seriously, it is an alternative of using weighted spatial lags.

Our method is applicable for researchers and practitioners as long as their data are naturally coded at some low administrative level, such as a municipality or a ZIP code. The method is most suited to be used under some fairly broad conditions:

1. Variables of the analysis that typically make sense at some spatial (regional) unit (for example, consider variables like the number of hospital beds, doctors, university classes).
2. The analysis involves processes that are not constrained within a single spatial unit, such as spatial spillover effects (for example, knowledge flows are diffused over space).

¹⁹ Also, the DtB method generates a variety of “shapes,” but importantly, they show that this feature does not matter.

²⁰ The method is simple: once the map is digitized, it is fast and may be modified easily. One additional difference is the speed of calculating robustness measures; our code is substantially faster. In lack of available codes, we coded the Briant et al. (2010) method ourselves. On a laptop, our code run was about 10–20 times faster.

3. Regions are not defined by first geography separators (for example, seas and mountains do not prevent transport and knowledge flows).
4. Building block spatial units are heterogeneous—and the average size of units is correlated with some unobserved feature (for example, the size of the Benelux and Spanish regions differs greatly and this difference is correlated with socioeconomic factors such as urbanization).

We believe that our method is an improvement on existing ones such as spatial lags (SL), DtB by [Briant et al. \(2010\)](#), added area under fixed radius circles (circles), while bringing together some advantages of these approaches. Our approach ends up comparing equally sized units thus avoiding bias from the endogeneity of spatial structure (i.e., concentration of small units in particular areas). This is true for the circles model, but both SL and DtB add units rather than distance. Importantly, our method counts all units only once and hence does not oversample dense areas with many small units (such as Circles). While it improves some SL models as well, it is less flexible in modeling nonlinear relationships within units. Apparently, it is a better method when spatial units are of different size in a non-random fashion. Other methods, for example [Briant et al. \(2010\)](#), disregard unit area size. In this respect, distance-weighted SL and circles are similar to our approach.

The advantages of our method have several practical applications. In particular, we showed that the size of artificial grid units does matter in the measurement of the agglomeration premium. Such information is useful when we aim to compare estimates calculated for various datasets in different parts of the world. We also showed that the method may be used to test the impact of policies by comparing actual and pseudo-administrative units. In our case, regional boundaries had no impact on the wage elasticity of density.

6 Appendix

See Table 3. See Figs. 9, 10, 11, 12, 13, 14 and 15.

Table 3 Summary of Hungarian administrative spatial units and grids

	<i>Number of units</i>	<i>Area km²</i>		
		Mean	Min	Max
Municipality	3125	29.55	0.71	483.22
15 km grid	440	210.37	1.52	607.21
Micro-region	150	620.23	103.08	1573.10
26 km grid	150	605.35	67.61	1146.01
39 km grid	75	1235.50	152.20	2060.82

Hungarian municipalities are NUTS5-level units, and micro-regions are NUTS4-level units
Figures for grids are averages over 300 replications

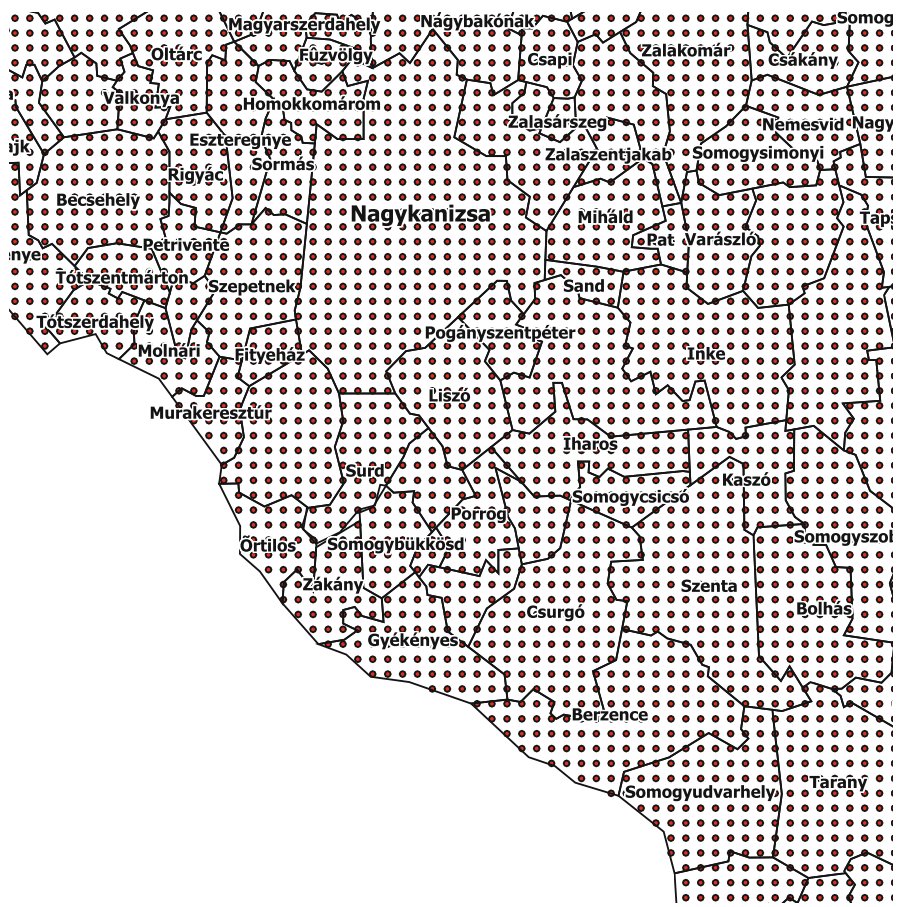


Fig. 9 Markers serve as basic elements to be shaken

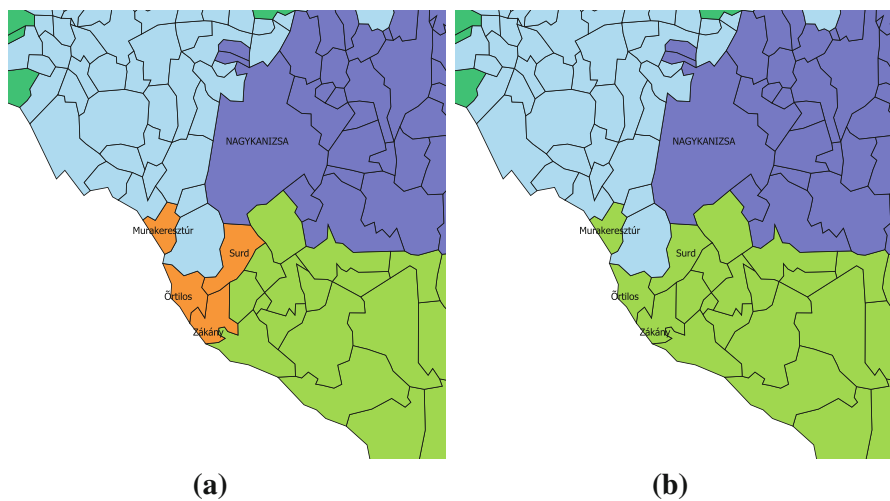


Fig. 10 Adjustment for small areas near the border: 26×26 km. **a** Before adjustment. **b** After adjustment

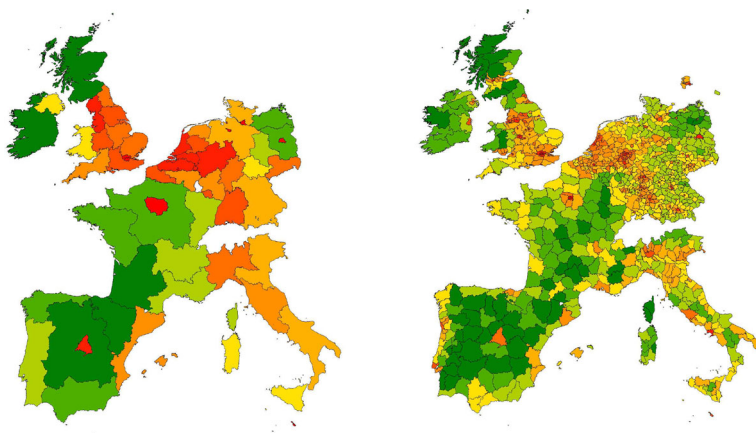


Fig. 11 Population density in European regions

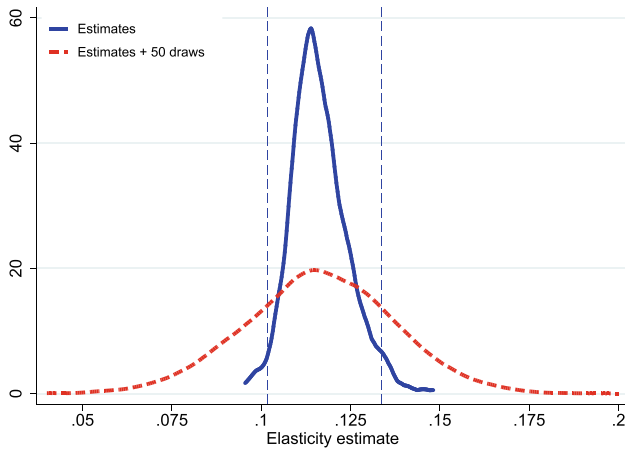


Fig. 12 Additional SD. This figure shows the distribution of estimated coefficients on population in Model 1—agglomeration elasticity of wage—(blue line) as well as an extended distribution (red dashed line). The extended distribution takes into account the uncertainty of each elasticity estimate: β_i^{km26} where $i = \{1 \dots 300\}$. For each, we randomly draw 50 coefficients from normal distribution $N(\beta_i^{km26}, \sigma_i^{km26})$. All together red dashed line is a density over 50×300 elasticity value

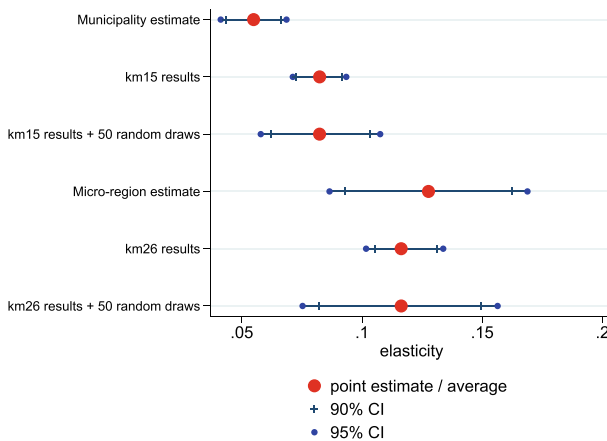


Fig. 13 Additional SD2. This figure shows the distribution boxplots for population coefficient estimates in Model 1 (agglomeration elasticity of wage) produced by various approaches. These are from top to bottom: (1) elasticity estimate using municipality-level data, confidence intervals are calculated using the standard deviation estimates of the elasticity. (2) Estimates on grids using km15 sizes, confidence interval is created using the deviation from the 300 estimates. (3) Estimates on grids using km15 sizes, confidence interval takes into account estimation uncertainty of each of the 300 estimates by adding 50 random normal draws: $N(\beta_i^{km15}, \sigma_i^{km15})$. (4) Elasticity estimate using micro-region-level data, confidence intervals are calculated using the standard deviation estimates of the elasticity. (5) Estimates on grids using km26 sizes, confidence interval is created using the deviation from the 300 estimates. (6) Estimates on grids using km26 sizes, confidence interval takes into account estimation uncertainty of each of the 300 estimates by adding 50 random normal draws: $N(\beta_i^{km26}, \sigma_i^{km26})$

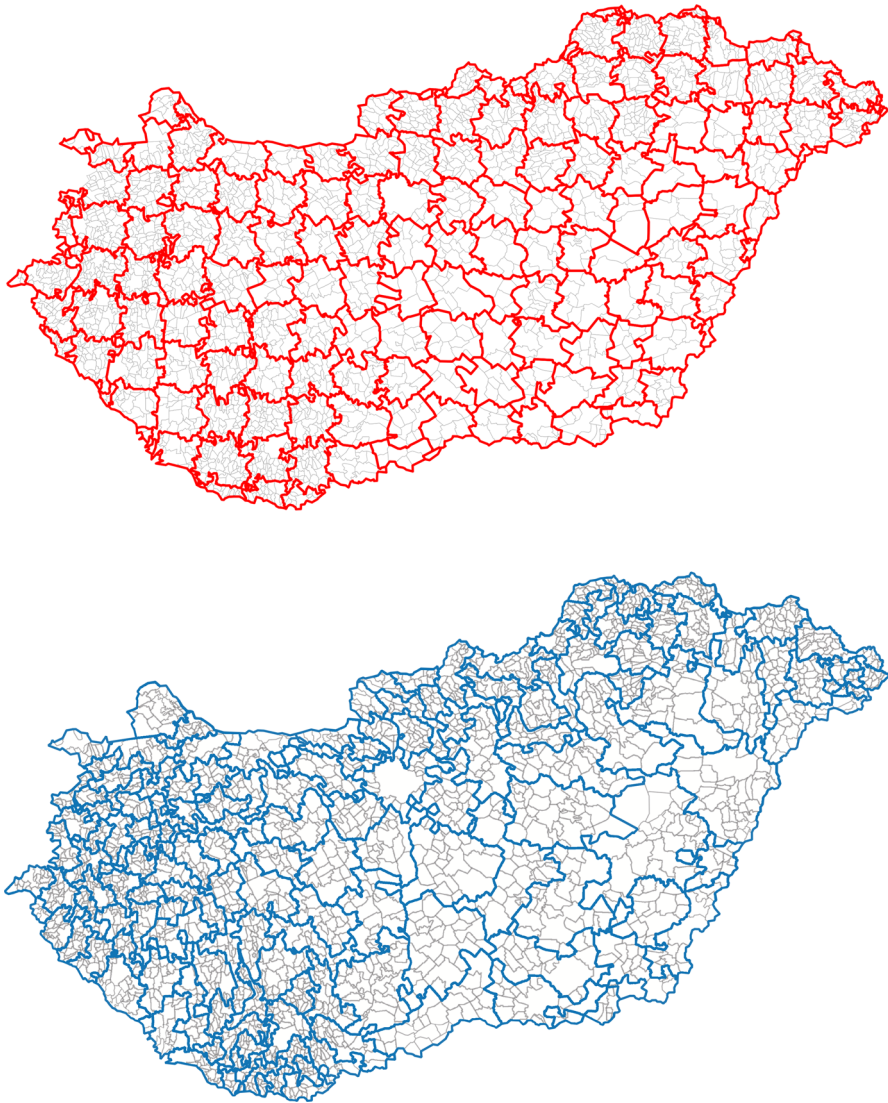


Fig. 14 A realization of 26×26 km units (above) and 150 seeded DtB (below)

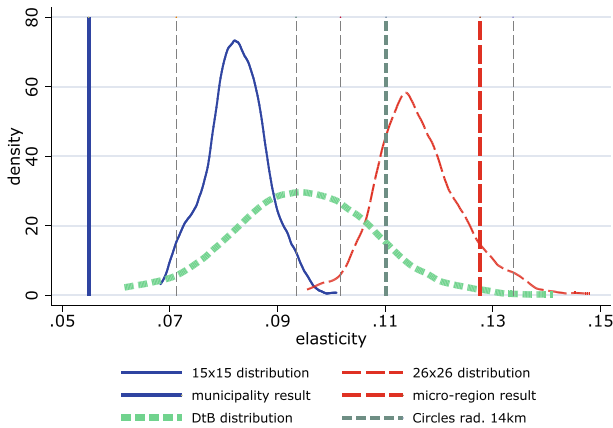


Fig. 15 Comparison to dots-to-boxes results. This figure shows the exact replication of Fig. 4, and it additionally displays DtB results too. It shows the coefficient estimate on population in Model 1 (agglomeration elasticity of wage). The thick blue (solid) line gives the point estimate for the municipality estimate. The thin blue (solid) line gives the distribution of point estimates over 300 point estimates on the 15×15 grid. The thick red (dashed) line gives the point estimate for the micro-region estimate. The thin red (dashed) line gives the distribution of point estimates over 300 point estimates on the 26×26 grid. The very thin dashed lines represent the 5% confidentiality thresholds. Additionally, the very thick mint-green (short dash) line shows the distribution of point estimates over 300 realizations of Dots-to-boxes spatial arrangements using 150 seeds

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